

★ WEEK 3 · PART 1 · PRACTICE

AR, MA, and Wold, Worked by Hand

Read a Series' Memory in Closed Form, Then Tell AR From MA

AR(1) properties, MA(1) cutoff, and the AR-MA duality

Part 1's building blocks, with the arithmetic shown. We compute an AR(1)'s mean, variance, and autocorrelation in closed form, then an MA(1)'s variance and its sharp ACF cutoff - and see how a stationary AR is secretly an infinite MA, the heart of Wold's theorem.

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PART 1 · APPLICATIONS

Memory, in Closed Form

Case A: an AR(1)'s mean, variance, and geometrically decaying autocorrelation, plus its stationarity condition and memory half-life. Case B: an MA(1)'s variance and sharp ACF cutoff, and the AR-MA duality that makes every stationary series an MA at heart. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

A series is white noise, filtered by memory

AR carries yesterday's value forward; MA carries yesterday's shock. Both build structure from pure noise.

2

AR autocorrelation decays; MA cuts off

An AR(1)'s ACF fades geometrically and never reaches zero; an MA(1)'s is a single spike, then exactly zero.

3

AR and MA are two sides of a coin

A stationary AR is an infinite MA and vice versa - Wold's theorem: every stationary series is an MA at heart.

Everything here is a calculator exercise: one AR in closed form, one MA cutoff, one duality.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from an AR or MA coefficient.
- Compute the closed-form properties.
- Then read the memory.

What you should watch

- How the AR autocorrelation decays.
- How the MA autocorrelation cuts off.
- How an AR becomes an infinite MA.

The AR(1) Memory

Today is a fraction of yesterday

A mean-reverting return follows an AR(1): today is a fraction of yesterday plus a fresh shock. That single coefficient fixes everything - the long-run mean, the variance, and how autocorrelation decays. We compute all three in closed form, confirm stationarity, and read off the memory's half-life

The model

- Coefficient ϕ of 0.6.
- Constant of 0.4, shock variance 1.0.
- A persistent, mean-reverting series.

What we'll do by hand

- Compute the mean and variance.
- Compute the autocorrelation at each lag.
- Find the half-life of the memory.

The AR(1) Model

A fraction of yesterday, plus a shock

$$y_t = c + \phi y_{t-1} + \varepsilon_t \quad (\text{today is a fraction } \phi \text{ of yesterday, plus a fresh shock})$$

The pieces

- Phi scales yesterday's value into today.
- The constant sets the long-run level.
- The shock is fresh white noise.

The intuition

- A high phi means a long memory.
- A low phi means the past fades fast.
- Below one keeps it stationary.

Mean and Variance

In closed form

$$\mu = \frac{c}{1-\phi} = \frac{0.4}{0.4} = \mathbf{1.0}, \quad \gamma_0 = \frac{\sigma^2}{1-\phi^2} = \frac{1.0}{0.64} = \mathbf{1.5625}$$

The mean

- The constant over one minus phi.
- 0.4 over 0.4 equals 1.0.
- That's the level it reverts to.

The variance

- The shock variance over one minus phi-squared.
- 1.0 over 0.64 equals 1.5625.
- Larger than the shock, because of persistence.

Autocorrelation and Stationarity

A geometric decay

$$\rho_k = \phi^k : 0.6, 0.36, 0.216, \dots \text{ (geometric decay); stationary } \iff |\phi| < 1$$

The decay

- The autocorrelation at lag k is ϕ to the k .
- 0.6, 0.36, 0.216, and so on.
- It fades but never quite reaches zero.

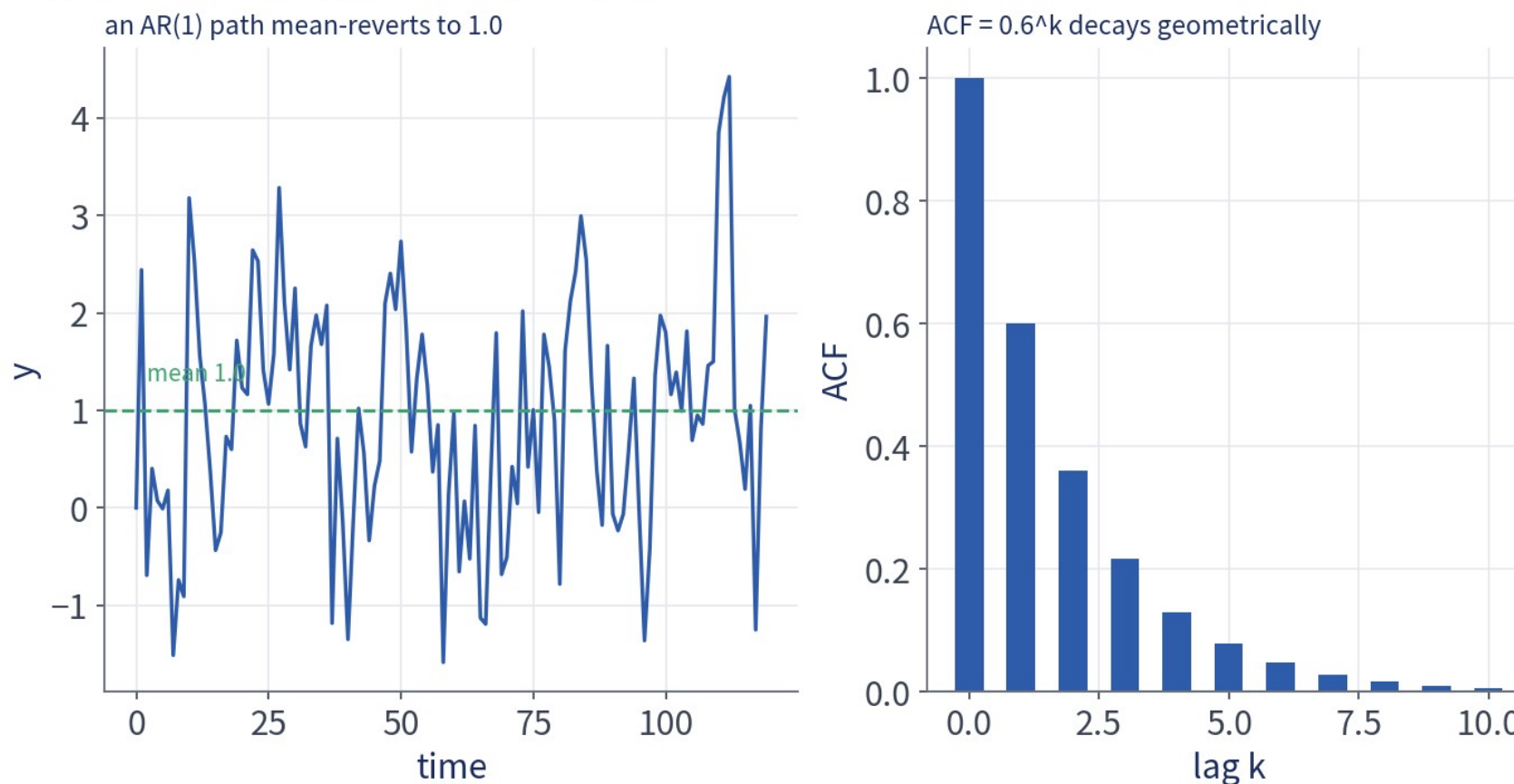
Half-life and stability

- The correlation halves in about 1.4 lags.
- Stationary because ϕ is below one.
- At one or above, it would never settle.

Persistence and Its Decay

On the left, an AR(1) path wanders but keeps returning to its mean of 1.0 - the pull of mean reversion. On the right, its autocorrelation function: a smooth geometric decay of 0.6 to the k , fading toward zero without ever reaching it. That never-quite-zero tail is the signature of an AR process.

Case A: AR(1) - persistent memory, geometrically decaying autocorrelation



Reading the Memory

What ϕ tells a trader

What ϕ encodes

- The strength of mean reversion.
- How fast a shock's effect fades.
- The half-life of predictability.

Why it matters

- A high ϕ means trends persist - momentum.
- A low or negative ϕ means quick reversal.
- The ACF's decay rate is the trading horizon.

MA(1) and the Duality

A finite echo, and Wold's coin

An MA(1) carries yesterday's shock, not yesterday's value - a finite echo. Its autocorrelation is a single spike, then exactly zero, a sharp contrast with the AR's slow decay. We compute the MA's variance and that cutoff, check invertibility, then reveal the duality: a stationary AR(1) is secretly an infinite MA - the essence of Wold's

or

The MA(1)

- Today's value is a shock plus part of yesterday's.
- Theta of 0.8, shock variance 1.0.
- A memory just one period long.

What we'll do by hand

- Compute its variance and lag-1 autocorrelation.
- Confirm the cutoff and invertibility.
- Write the AR(1) as an infinite MA.

The MA(1) Model

A finite echo of one shock

$$\gamma_0 = \sigma^2(1 + \theta^2) = \mathbf{1.64}, \quad \rho_1 = \frac{\theta}{1+\theta^2} = \frac{0.8}{1.64} = \mathbf{0.49}, \quad \rho_{k \geq 2} = \mathbf{0}$$

The variance

- The shock variance times one plus theta-squared.
- One times 1.64, so 1.64.
- Inflated by the echoed shock.

The cutoff

- Lag-1 autocorrelation is 0.49.
- Every lag beyond is exactly zero.
- A one-period memory, sharply bounded.

The Sharp Cutoff

And invertibility

lag k	1	2	3	4	5
MA(1) ACF	0.49	0	0	0	0
AR(1) ACF	0.60	0.36	0.22	0.13	0.08

The MA(1) autocorrelation is 0.49 at lag one and exactly zero everywhere after - a sharp cutoff. The AR(1) beneath it decays but never dies. That contrast is what identifies one from the other. The MA is invertible because θ is below one.

AR Equals Infinite MA

Wold's theorem, concretely

$$\text{AR}(1) \ \phi=0.6 \ = \ \text{MA}(\infty) : \ y_t = \sum_{j \geq 0} \phi^j \varepsilon_{t-j} = \varepsilon_t + 0.6\varepsilon_{t-1} + 0.36\varepsilon_{t-2} + \dots$$

The expansion

- Substitute the AR(1) into itself repeatedly.
- Each shock enters with weight ϕ to its age.
- 1, 0.6, 0.36, 0.216, fading away.

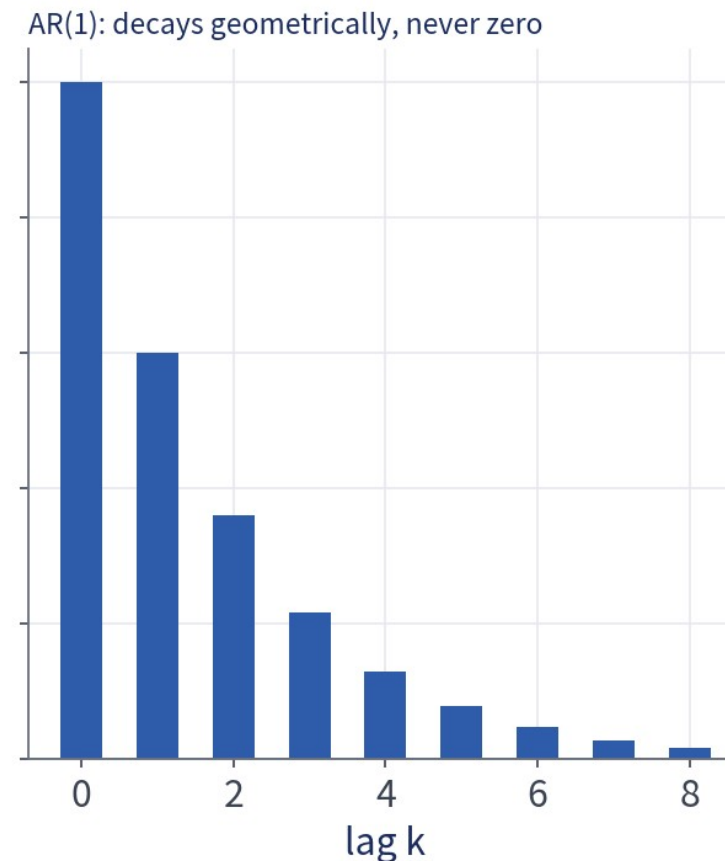
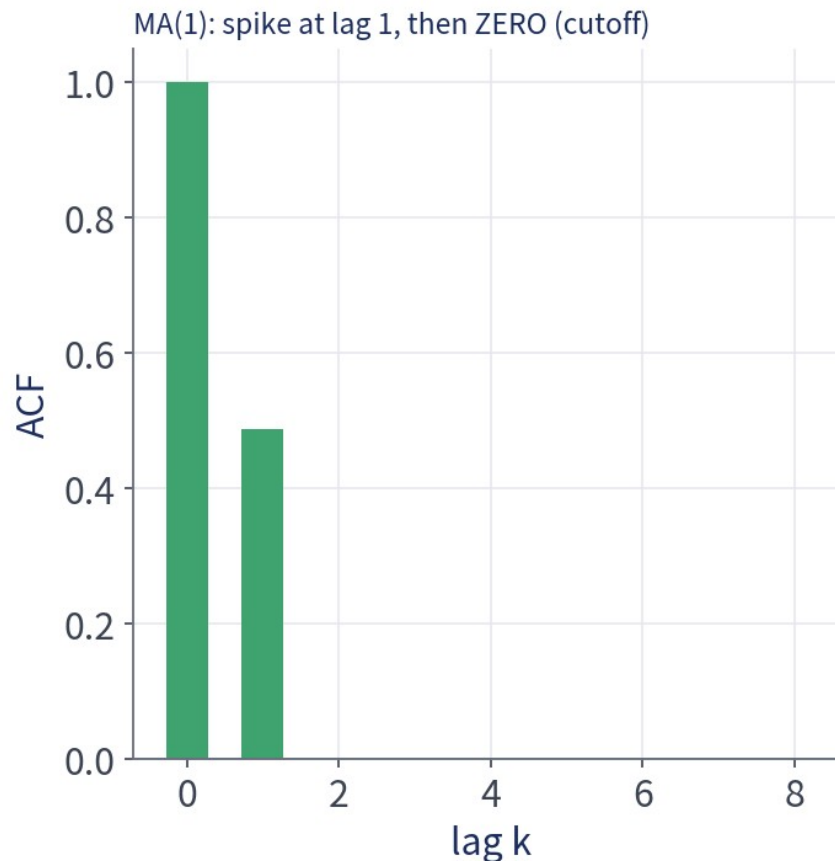
Why it matters

- A stationary AR is an infinite MA.
- Wold: every stationary series is an MA at heart.
- AR and MA are two views of one thing.

Cutoff Versus Decay

Side by side, the two signatures. The MA(1) autocorrelation (left) spikes at lag one and drops to exactly zero - a clean cutoff. The AR(1) (right) decays geometrically, tapering but never vanishing. Reading which pattern a series shows is how you tell an MA from an AR - the whole basis of identification.

Case B: cutoff vs decay - the signature that tells MA from AR



The Signature That Identifies

Cutoff, decay, and Wold

The identification rule

- A sharp ACF cutoff points to MA.
- A slow ACF decay points to AR.
- Next week turns this into a full method.

Why Wold matters

- Any stationary series has an MA form.
- So MA models are universal building blocks.
- AR is just a compact way to write an infinite MA.

The Two Tools

Memory, persistence, and correlation

AR and MA are the two atoms of linear time series. AR builds persistence from past values, giving an autocorrelation that decays; MA builds a finite echo from past shocks, giving one that cuts off. Their closed-form properties - mean, variance, autocorrelation - follow from a single coefficient each. And the duality binds them:

ie

AR(1) (Case A)

- Mean 1.0, variance 1.5625.
- ACF 0.6 to the k - geometric decay.
- Half-life about 1.4 lags.

MA(1) and duality (Case B)

- Variance 1.64, ACF 0.49 then zero.
- AR(1) equals an infinite MA.
- Wold: every stationary series is MA.

Reproduce the Properties and the Duality

Direct Claude Code to check your arithmetic

GOAL Compute AR(1) and MA(1) properties, then expand the AR as an infinite MA.

[INPUT] Prompt · Claude Code

```
Write and run part1_practice.py that:  
CASE A - AR(1), phi=0.6, c=0.4, sigma^2=1:  
  * print the mean, variance, and ACF for lags 1-5;  
  * print the memory half-life.  
CASE B - MA(1), theta=0.8:  
  * print the variance and the lag-1 ACF;  
  * confirm ACF is zero for lag >= 2;  
  * expand AR(1) as an MA and print psi_0..psi_4.  
Then plot the AR path with its ACF, and the MA-vs-AR  
ACF comparison.
```

Check against the slides

- AR mean 1.0, variance 1.5625.
- AR ACF 0.6, 0.36, 0.216, ...
- MA variance 1.64, ACF 0.49.
- $\psi_j = 0.6^j$.

Deliverable: part1_practice.py whose printed numbers match the AR and MA properties.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How does an AR(1)'s autocorrelation behave?

It decays geometrically as ϕ to the k

Q2 What's distinctive about an MA(1)'s ACF?

A single spike at lag one, then exactly zero

Q3 What does Wold's theorem say?

Every stationary series is an infinite MA at heart

All three come from the AR closed form, the MA cutoff, and the AR-MA duality.

Three Pitfalls to Avoid

Where AR and MA trip people

Common mistakes

- "An AR autocorrelation eventually hits zero."
- "MA and AR ACFs look the same."
- "Any ϕ gives a stationary AR."

The reality

- It decays toward zero but never exactly reaches it.
- MA cuts off sharply; AR decays - that's the tell.
- Stationarity needs ϕ below one in magnitude.

Two Calculations to Remember

Part 1 applications in one screen

1

AR(1) properties in closed form

With ϕ of 0.6, the mean was 1.0, the variance 1.5625, and the autocorrelation 0.6 to the k - 0.6, 0.36, 0.216 - a geometric decay with a half-life near 1.4 lags. One coefficient fixes the entire memory.

2

MA(1) cuts off; AR is an infinite MA

The MA(1) had variance 1.64 and a lag-1 autocorrelation of 0.49, then exactly zero - a sharp cutoff. And the AR(1) expanded into an infinite MA with weights 1, 0.6, 0.36, illustrating Wold: every stationary series is an MA at heart.

3

Cutoff versus decay is the tell

A sharp ACF cutoff signals MA; a slow decay signals AR. That single contrast, plus the AR-MA duality, is the foundation the next part builds identification on.

Next — Part 2 practice: turning the ACF and PACF signatures into a full ARMA identification method.